

Huffman Coding Algorithm :-

Huffman coding Algorithm is a lossless data compression algorithm. In computer science and information theory a Huffman code is a particular type of optimal prefix code that is commonly used for image and video compression. The Complexity of Huffman coding algorithm is $O(n \log n)$.

where n is the number of unique characters or symbols in the input data. This complexity arises from building the Huffman tree by repeatedly merging nodes and constructing the variable-length codes.

Applications of Huffman coding Algorithm :-

1. Used in image and video compression techniques like JPEG and MPEG.
2. Enabling efficient storage and transmission of multimedia content.
3. Search Engines: Huffman Coding is employed by search engines to optimize the storage and retrieval of data, improving search speeds.
4. Used in conventional compression formats like GZIP, BZIP2, PKZIP, etc.